Interdisciplinary Schools

NeuroPlay Workshop Homework

Visualizing Brain Signals (EEG) with MATLAB

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Homework Instructions

Important note: right after running the code for each question, take a full screenshot of your Live Script workspace (including the code you wrote and the resulting plot) and paste it into the "Screenshot Box" provided below that question in this same file.

- Every plot must include a title, axis labels (xlabel/ylabel), and a legend where needed.
- Wherever a question asks you to "comment on" something, write your answer as a comment (%) above that part of the code.
- Using custom colors, markers, and LineWidth to make your plots look nicer earns extra credit.

Question 1 — Comparing Brain Activity with Bar and Box Plots

Level: Medium

Suppose the brain activity of 30 subjects was recorded in three states: "Rest", "Task A", and "Task B". Simulate this data using the randn command as shown in the workshop (each group should have a different mean).

a) Plot the average activity of all three groups with a bar chart. Show the group names on the x-axis using set(gca,'XTickLabel',...).

b) Show the full distribution of all three groups' data with a boxplot in a separate figure, and set the group labels with 'Labels'.

c) Looking at the boxplot, write down which group has the greatest spread (IQR)? Write your answer as one comment line below the code.

Hint for This Question

- To simulate three groups with different means, use a pattern like this:

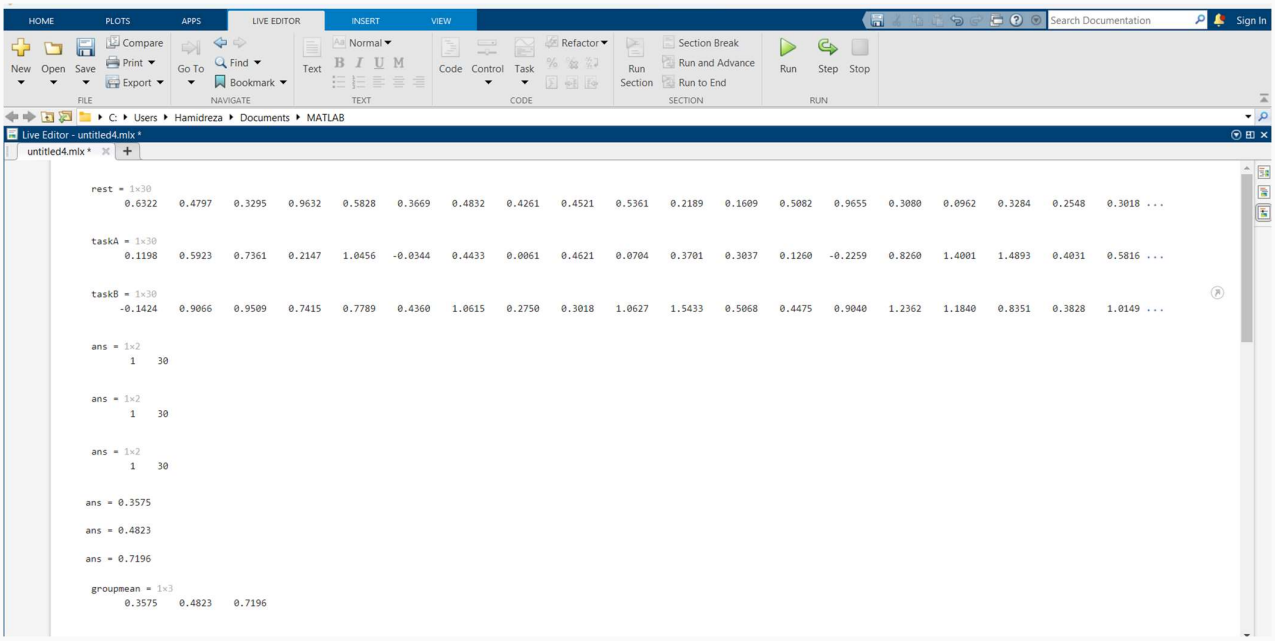
```
rest  = randn(1,30)*0.3 + 0.4;
taskA = randn(1,30)*0.4 + 0.6;
taskB = randn(1,30)*0.5 + 0.7;
```

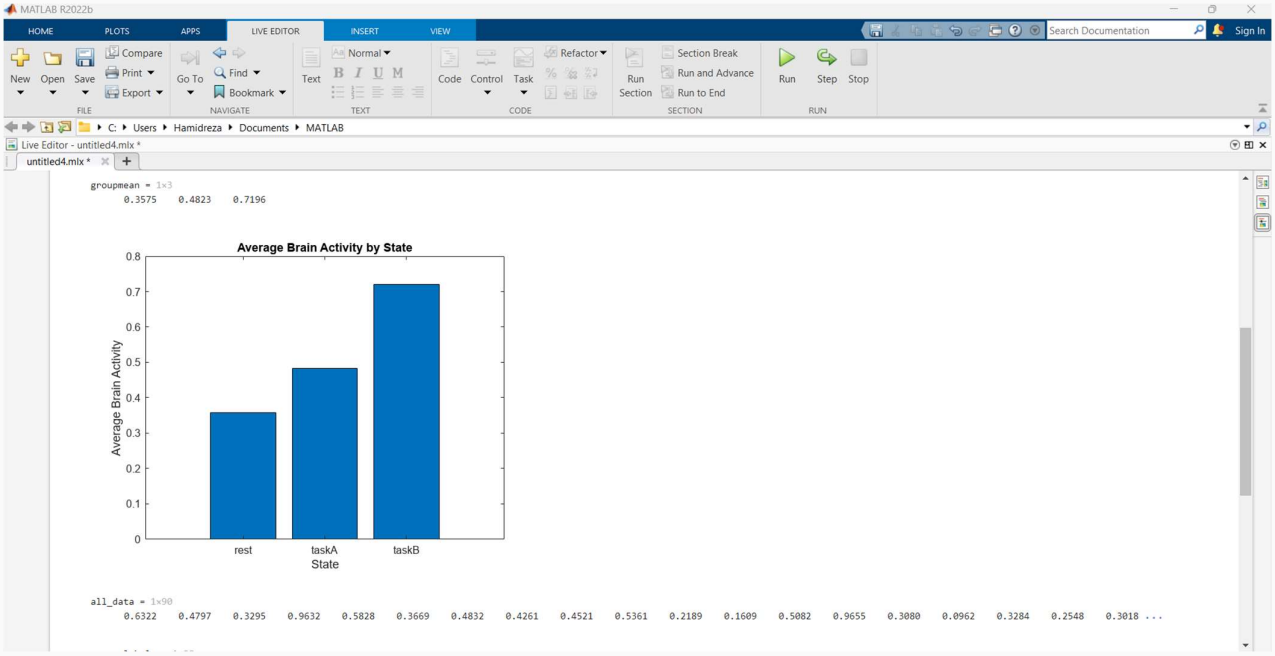
- For the bar chart, first compute each group's mean with mean() and pass a 3-element vector to bar
- For the boxplot, concatenate the three groups' data and group them with a numeric label vector (1, 2, 3)
- For part (c): in a boxplot, the taller the box (the distance between the 1st and 3rd quartile), the greater that group's spread (IQR) — you don't need an exact number, just compare visually.

Screenshot Box — Question 1

Paste your Live Script workspace screenshot (code + plot output) here

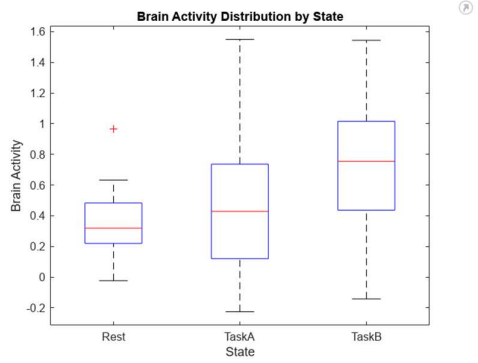
```
rest=randn(1,30)*0.3+0.4
taskA=randn(1,30)*0.4+0.6
taskB=randn(1,30)*0.5+0.7
size(rest)
size(taskA)
size(taskB)
mean(rest)
mean(taskA)
mean(taskB)
groupmean=[mean(rest) mean(taskA) mean(taskB)]
bar(groupmean)
set(gca,'XTickLabel',{'rest','taskA','taskB'})
title('Average Brain Activity by State')
xlabel('State')
ylabel('Average Brain Activity')
figure
all_data=[rest taskA taskB]
group_labels = [ones(1,30) 2*ones(1,30) 3*ones(1,30)]
boxplot(all_data,group_labels,'labels',{'Rest','TaskA','TaskB'})
title('Brain Activity Distribution by State')
xlabel('State')
ylabel('Brain Activity')
% The group with the largest Q3 - Q1 has the greatest spread (IQR).
```


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TEXT
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Task
Refactor
CODE
Run Section
Section Break
Run and Advance
Run to End
SECTION
Run
Step
Stop
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Live Editor - untitled4.mlx *
untitled4.mlx *
rest = 1×30
0.6322 0.4797 0.3295 0.9632 0.5828 0.3669 0.4832 0.4261 0.4521 0.5361 0.2189 0.1609 0.5082 0.9655 0.3080 0.0962 0.3284 0.2548 0.3018 ...
taskA = 1×30
0.1198 0.5923 0.7361 0.2147 1.0456 -0.0344 0.4433 0.0061 0.4621 0.0704 0.3701 0.3037 0.1260 -0.2259 0.8260 1.4001 1.4893 0.4031 0.5816 ...
taskB = 1×30
-0.1424 0.9066 0.9509 0.7415 0.7789 0.4360 1.0615 0.2750 0.3018 1.0627 1.5433 0.5068 0.4475 0.9040 1.2362 1.1840 0.8351 0.3828 1.0149 ...
ans = 1×2
1 30
ans = 1×2
1 30
ans = 1×2
1 30
ans = 0.3575
ans = 0.4823
ans = 0.7196
groupmean = 1×3
0.3575 0.4823 0.7196


MATLAB R2022b
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Live Editor - untitled4.mlx *
untitled4.mlx *
groupmean = 1×3
0.3575 0.4823 0.7196
Average Brain Activity by State
Average Brain Activity
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0
rest
taskA
taskB
State
all_data = 1×90
0.6322 0.4797 0.3295 0.9632 0.5828 0.3669 0.4832 0.4261 0.4521 0.5361 0.2189 0.1609 0.5082 0.9655 0.3080 0.0962 0.3284 0.2548 0.3018 ...

```
all_data = 1x90
    0.6322    0.4797    0.3295    0.9632    0.5828    0.3669    0.4832    0.4261    0.4521    0.5361    0.2189    0.1609    0.5082    0.9655    0.3080    0.0962    0.3284    0.2548    0.3018 ...

group_labels = 1x90
     1     1     1     1     1     1     1     1     1     1     1     1     1     1     1     1     1     1     1     1     1     1     1     1     1     1     1     2     2 ...
```



Question 2 — Displaying Frequency Band Power as Percentages

Level: Easy

An individual's EEG signal power was measured in four frequency bands as follows: Delta = 18, Theta = 27, Alpha = 40, Beta = 15 (arbitrary units).

a) Plot these four values with pie, using each band's name as a label (names).

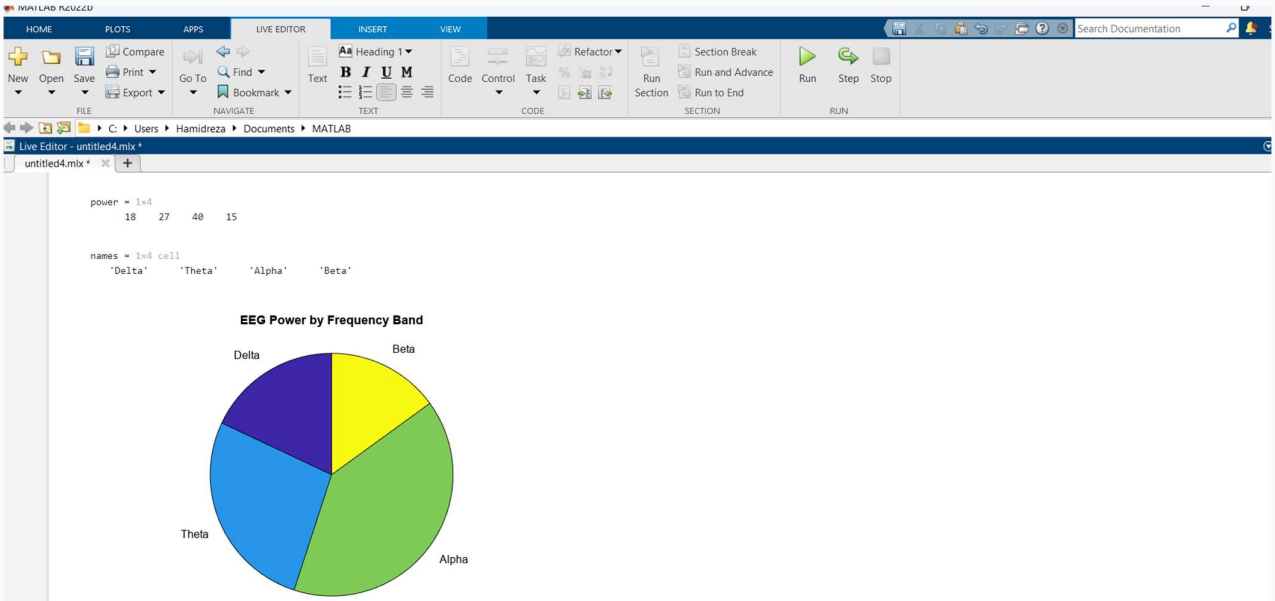
b) Plot the same data again with pie3, and in one comment line write which one is better suited for a presentation slide, and why.

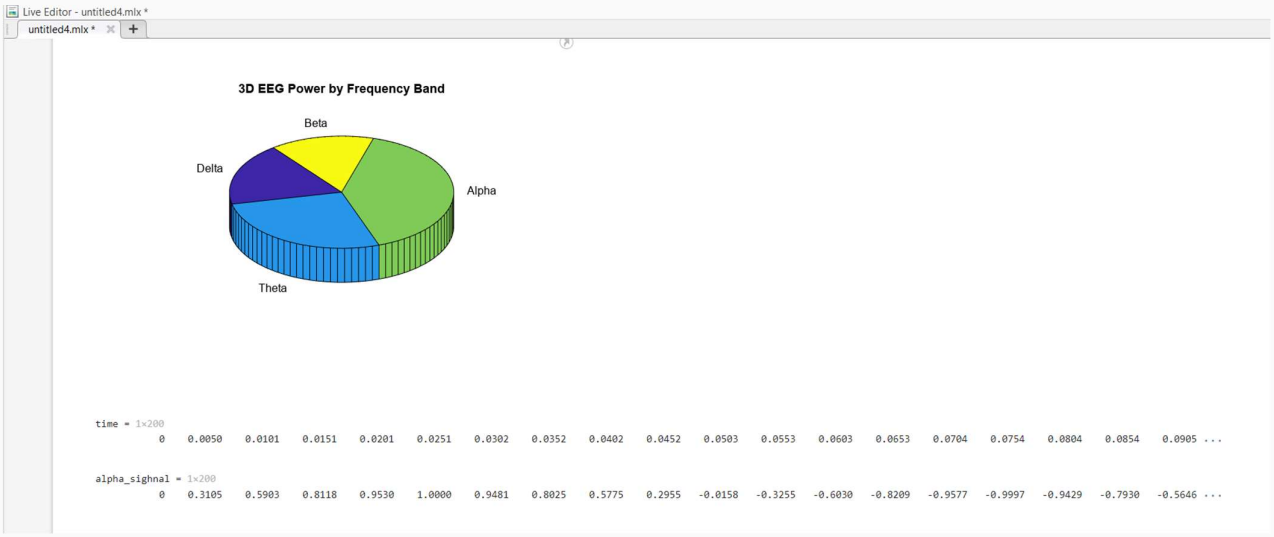
c) Simulate an alpha signal with a frequency of 10 Hz and 200 samples (sin(2*pi*10*time)) and plot a histogram of its amplitude; does the distribution look normal or not? (one comment line).

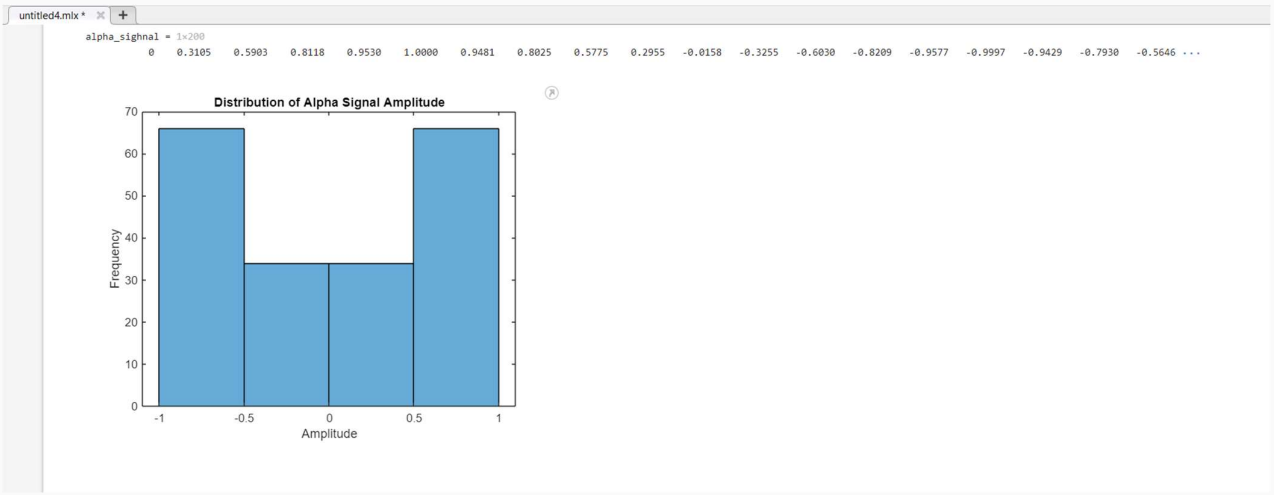
Screenshot Box — Question 2

Paste your Live Script workspace screenshot (code + plot output) here

```
power=[18 27 40 15]
names={'Delta','Theta','Alpha','Beta'}
pie(power,names)
title('EEG Power by Frequency Band')
figure
pie3(power,names ...
    )
title('3D EEG Power by Frequency Band')
% The 2D pie chart is better for presentation slides because it is clearer and easier to compare.
time=linspace(0,1,200)
alpha_sighnal=sin(2*pi*10*time)
figure
histogram(alpha_sighnal)
title('Distribution of Alpha Signal Amplitude')
xlabel('Amplitude')
ylabel('Frequency')
% The amplitude distribution is not normal because it is concentrated near the peaks of the sine wave.
```







Question 3 — Screening Brain Channels with Logical Operators

Level: Medium

Generate a vector activity containing the activity of 8 brain channels using activity = rand(1,8), then:

a) Use the > operator to identify channels with activity above 0.6.

b) Use the & operator to find channels that are simultaneously between 0.3 and 0.6.

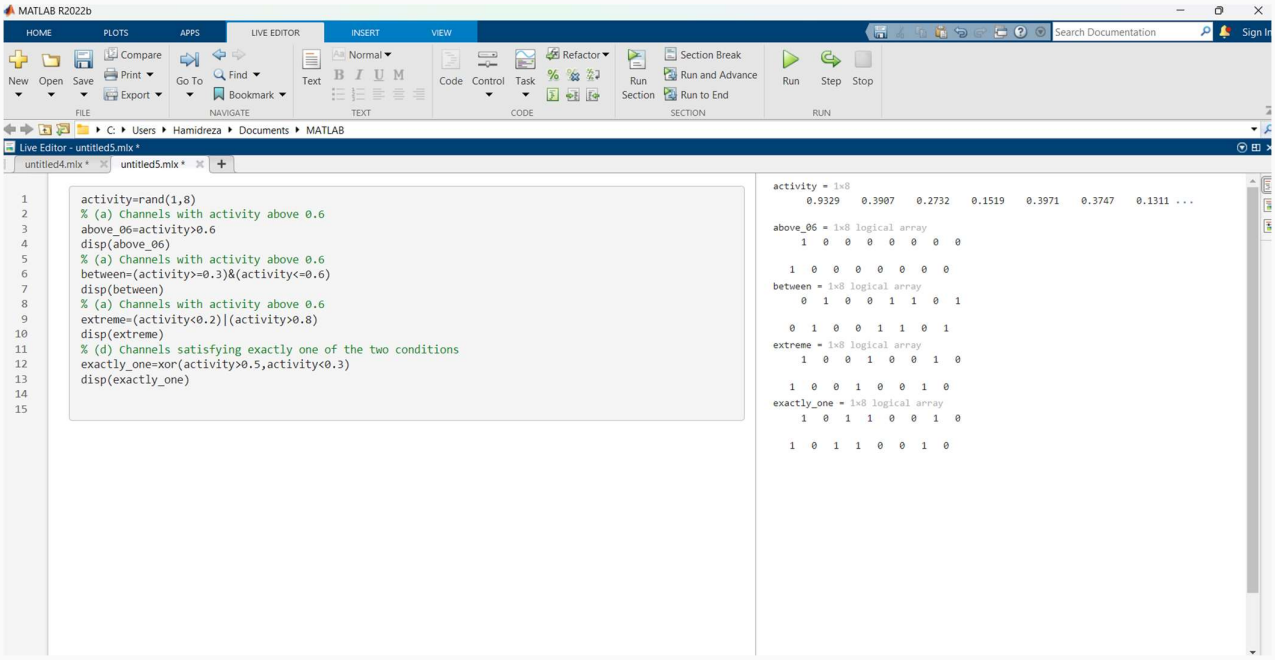
c) Use the | operator to find channels that are either very low (below 0.2) or very high (above 0.8).

d) Use the xor function to identify channels that satisfy “exactly one” of the two conditions (activity > 0.5) or (activity < 0.3).

- Print the result of each of the four steps with disp so the logical output (0 and 1) for each channel is visible.

Screenshot Box — Question 3

Paste your Live Script workspace screenshot (code + plot output) here



Question 4 — Detecting and Displaying Outliers in a Brain Signal

Level: Challenge

This question combines “plotting” and “logical operators”. Build a noisy EEG signal with a frequency of 10 Hz and a length of 2 seconds:

```
time = 0:0.01:2;  signal = sin(2*pi*10*time) + 0.3*randn(size(time));
```

a) Using logical operators, identify the samples of the signal whose amplitude is greater than mean(signal)+2*std(signal) or less than mean(signal)-2*std(signal) as outliers (the output of this step should be a logical vector).

b) First plot the original signal with plot (thin blue line); then, using hold on, mark the outlier points found in part (a) on the same plot with scatter (red color, '*' star marker, larger size).

c) Fully set the title, axis labels, and legend (e.g., 'EEG signal' and 'Outliers').

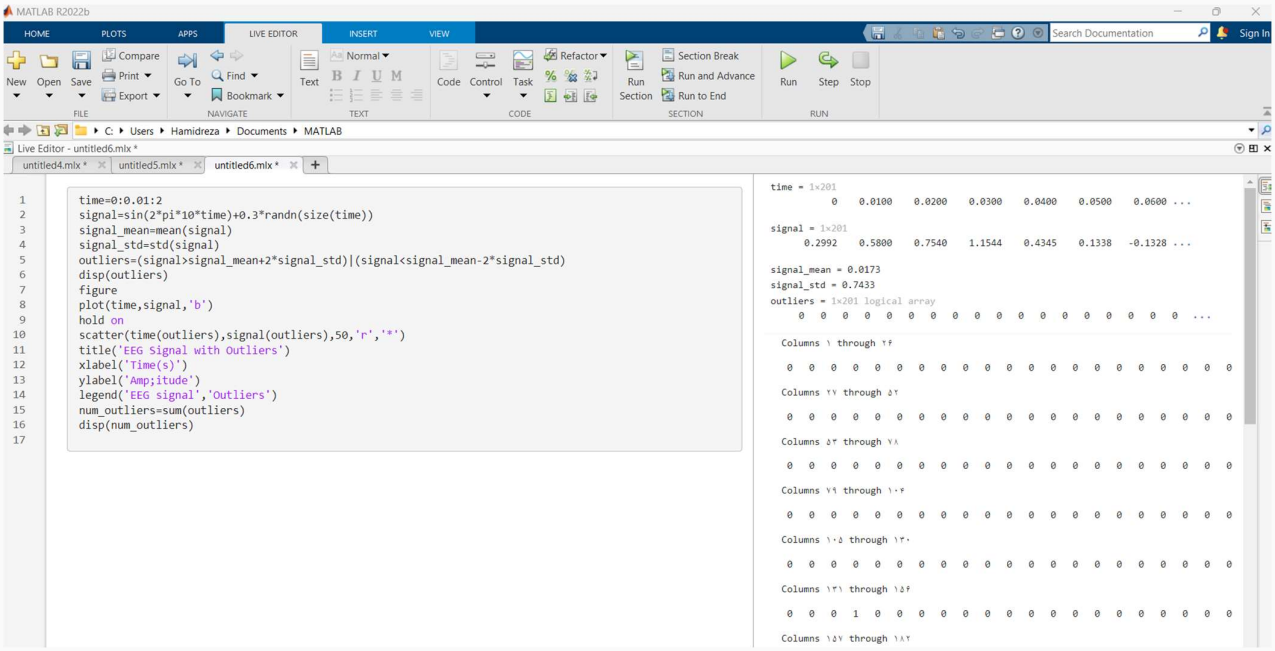
d) Using the sum function on the logical vector from part (a), compute and print the total number of outliers found with disp.

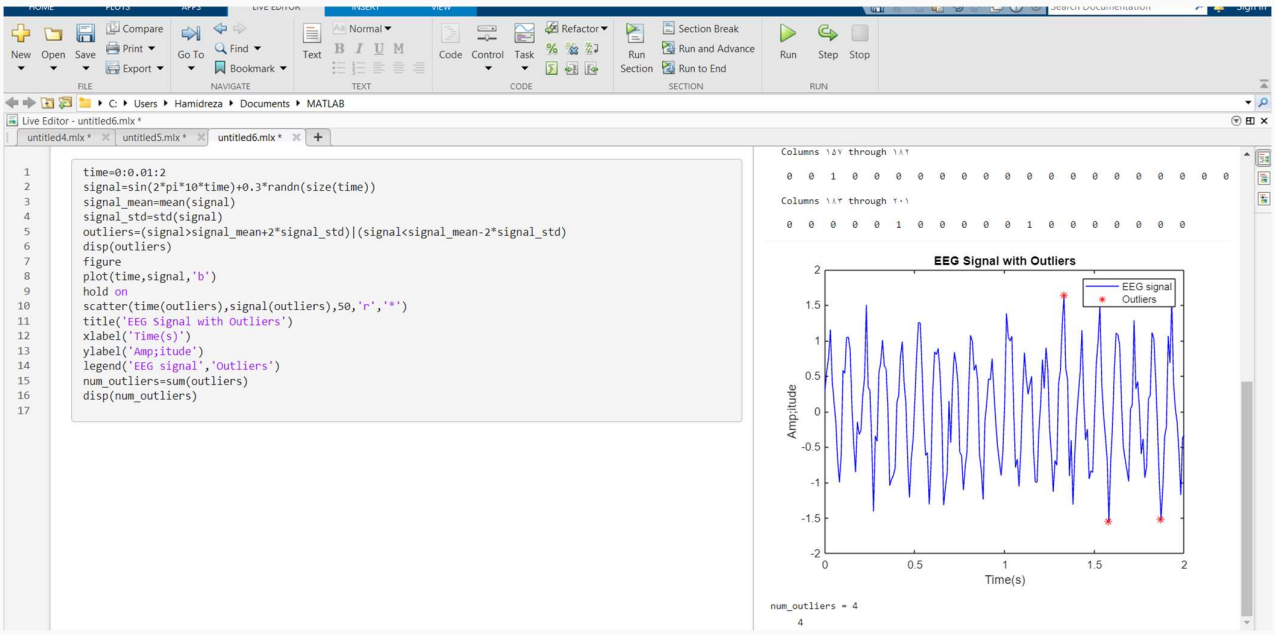
Hint for This Question

- First store the mean and standard deviation of the signal in separate variables so the code is more readable:
- Then build the logical outlier vector using the | (OR) operator between the two conditions:
- For plotting, first draw the full signal, then add just the outlier points on the same plot (remember you can index directly into time and signal with outliers):
- To count the outliers, simply apply sum to the logical vector (not to the signal values):

Screenshot Box — Question 4

Paste your Live Script workspace screenshot (code + plot output) here





Good luck! For each question, place your code + plot output (screenshot) in this same file.